D. Don't fear, DravDe is kind

time limit per test: 2 seconds
memory limit per test: 256 megabytes

A motorcade of n trucks, driving from city «Z» to city «3», has approached a tunnel, known as Tunnel of Horror. Among truck drivers there were rumours about monster DravDe, who hunts for drivers in that tunnel. Some drivers fear to go first, others - to be the last, but let's consider the general case. Each truck is described with four numbers:

- v — value of the truck, of its passangers and cargo
- c — amount of passanger on the truck, the driver included
- l — total amount of people that should go into the tunnel before this truck, so that the driver can overcome his fear («if the monster appears in front of the motorcade, he'll eat them first»)
- r — total amount of people that should follow this truck, so that the driver can overcome his fear («if the monster appears behind the motorcade, he'll eat them first»).

Since the road is narrow, it's impossible to escape DravDe, if he appears from one side. Moreover, the motorcade can't be rearranged. The order of the trucks can't be changed, but it's possible to take any truck out of the motorcade, and leave it near the tunnel for an indefinite period. You, as the head of the motorcade, should remove some of the trucks so, that the rest of the motorcade can move into the tunnel and the total amount of the left trucks' values is maximal.

Input

The first input line contains integer number n $(1 \le n \le 10^5)$ — amount of trucks in the motorcade. The following n lines contain four integers each. Numbers in the i-th line: v_i, c_i, l_i, r_i $(1 \le v_i \le 10^4, 1 \le c_i \le 10^5, 0 \le l_i, r_i \le 10^5)$ — describe the i-th truck. The trucks are numbered from 1, counting from the front of the motorcade.

Output

In the first line output number k — amount of trucks that will drive into the tunnel. In the second line output k numbers — indexes of these trucks in ascending order. Don't forget please that you are not allowed to change the order of trucks. If the answer is not unique, output any.

Examples

input Copy

```
5
1 1 0 3
1 1 1 2
1 1 2 1
1 1 3 0
2 1 3 0
```

output Copy

```
4
1 2 3 5
```

input Copy

```
5
1 1 0 3
10 1 2 1
2 2 1 1
10 1 1 2
3 1 3 0
```

output Copy

```
3
1 3 5
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 28 (Codeforces format)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
372689381	Apr/27/2026 12:39	Accepted
372689295	Apr/27/2026 12:38	Accepted
372689177	Apr/27/2026 12:36	Accepted

372689094	Apr/27/2026 12:35	**Accepted**
372688891	Apr/27/2026 12:32	Time limit exceeded on test 1
372688648	Apr/27/2026 12:29	Time limit exceeded on test 49

→ Problem tags

binary search | data structures | dp | hashing | *2400

No tag edit access

→ Contest materials

- Announcement
- Tutorial #1 (en)
- Tutorial #2
- Tutorial #3 (ru)
- Tutorial #4
- Tutorial #5

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